INCH--POUND
MIL-PRF-45333E(AT)
29 November 1996
SUPERSEDING
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29 June 1979

PERFORMANCE SPECIFICATION

TRAILER, FLAT-BED: 10 TON, 4 WHEEL M345

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a tactical flat-bed, 10 ton, 4 wheel type trailer designated as M345.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Other Government documents, drawings, and publications. The following other Governments documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2330
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NUCLEAR REGULATORY COMMISSION (NRC)

Code of Federal Regulations (CFR) - Title 10, Parts 30 and 40.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available form the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

DRAWINGS

ARMY

8358914 - Trailer, Flatbed: 10 Ton, 4 Wheel, M345.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for inspection by Attributes (DoD Adopted).

(Applications for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

2.3 Order of precedence. In the event of a conflict between the text of this document and references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

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3.2 Material. Material shall be capable of meeting all of the operational and environmental requirements specified herein (see 4.5.1). Recovered materials shall be used to the maximum extent practicable. Materials and special treatments shall be utilized to prevent corrosion and to resist harmful effects of dust, water and fungi. Dissimilar metals shall not be used in intimate contact without protection from galvanic corrosion.

3.2.1 Asbestos, cadmium, and radioactive materials. Asbestos, cadmium, and radioactive materials shall not be used in this item. Radioactive material is defined by Title 10, Code of Federal Regulations, Parts 30 and 40, and other radioactive material in which radioactivity is greater than 0.002 microcuries per gram or 0.01 microcuries total activity for the item.

3.3 Design and construction. Trailer components, subassemblies and assemblies shall be fabricated and assembled into a complete trailer in accordance to the manufacturer's design and construction except for the pertinent items on drawing 8358914, which affect performance, function, envelope dimensions, and interface.

3.3.1 Seals. When fording or operating in mud, sand or snow, seals shall prevent entrance of foreign matter into the bearings, which are exposed to contamination during these operations. All bearings seals shall restrict the leaking of lubricants from bearings (see 4.5.2).

3.3.2 Electrical system. Electrical circuits shall maintain continuity from end to end without evidence of internal or external shorts during all trailer operating conditions. All stop, signal, and breaking lights shall be of the standard commercial lighting for trailers (see 4.5.2).

3.3.3 Tires and inner tubes. Tires shall be cross country tires with tire flaps. Inner tubes shall be of the proper type and size for trailer applications (see 4.5.2).

3.3.4 Adjustable mechanisms. All adjustments mechanisms shall function properly and minimum adjustments settings during all vehicle operating conditions (see 4.5.2).

3.4 Performance. Trailer shall meet performance requirements of the specification when equipped as specified herein, serviced with standard products, loaded with rated payload of 26 000 pounds (lb), and coupled to the tow vehicle. The trailer, serviced and equipped for existing climate conditions, shall operate as specified without extra equipment.

3.4.1 Environment. The trailer shall operate under extreme conditions of weather and altitude in ambient air temperatures ranging from -65 to + 115 degrees Fahrenheit(°F). The complete trailer when in storage shall withstand a temperature of -80°F, without deterioration that may cause failure of any component (see 4.5.3.1).

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3.4.2 Highway (paved) and gravel roads, and speeds. The trailer shall be loaded to a 26 000 payload, or with a simulation load of equal weight, uniformly distributed over the load area and shall be towed at an average speed of 50 miles per hour (mph) over relatively level, smooth, improved and pre-paved hand-surfaced roads, and a speed of 20 mph over gravel roads, without damage to the trailer tow vehicle (see 4.5.3.2). For cross country, the payload shall be 20 000 lb at an average speed of 20 mph. The trailer shall track the tow vehicle in a straight line on a level, smooth, paved surface without deviating more then 3 inches (in.) to either side of the path of the tow vehicle.

3.4.3 Cross-country payload and speeds. The trailer, loaded with cross country rated payload of 20 000 lb uniformly distributed over the load area, shall be capable of being towed over unimproved roads, trails, open fields, rolling hills and cross country terrain at average speeds of 20 mph without damage to trailer or tow vehicle (see 4.5.3.3).

3.4.4 Grade and slope operation. The trailer shall operate without slipping or upsetting when ascending or descending longitudinal grades up to 40 percent(%) and side slopes up to 20% (see 4.5.3.4).

3.4.5 Service brakes. The combined service brakes of the trailer and the tow vehicle shall, under all conditions of loading, bring the combination to a positive stop within a distance of 30 feet (ft) from a speed of 20 mph on dry, smooth and level road, free of loose material (see 4.5.3.5).

3.4.5.1 Service breaking system. The trailer air reservoir shall be charged with air from 100 to 105 pounds per square inch (psi). After disconnection of the brake air supply line, service brakes shall be applied to all wheels of the trailer, not display fluid leakage, and maintain air pressure with a drop of no less than 0.5 psi per minute and no more than 5 psi per minute (see 4.5.3.5.1).

3.4.5.2 Automatic actuation. The trailer shall be equipped with an automatic actuating device to apply the trailer brakes upon breakaway from the tow vehicle. When loaded to a 26 000 lb highway payload, the automatic actuating device shall maintain application of the brakes and hold the trailer stationary on a 20% grade for not less than 15 minutes (see 4.5.3.5.2).

3.4.6 Fording ability. When towed through fresh or salt water to a depth great enough to entirely submerge the chassis for periods up to 15 minutes duration, vehicle or its components shall sustain no damage. This shall be accomplished without the use of special kits and with a minimum of preparation or servicing before and after fording operations (see 4.5.3.6).

3.4.7 Landing legs. Each landing leg shall raise, support and hold its portion of the load and shall withstand the strain imposed, without damage, and retract; and when retracted, shall clear the ground in cross country operation (see 4.5.3.7).

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3.4.8 Turning ability. When coupled to the tow vehicle and operating in a minimum turning circle, the trailer shall follow without cramping, damage or interference between the trailer and tow vehicle (see 4.5.3.8).

3.4.9 Wheel and tire balancing. Each wheel assembly, tube and tire shall be balanced to preclude wheel tramp and drumming. At road speeds up to 50 mph, each wheel assembly, tube and tire shall be checked for unusual noise (see 4.5.3.9).

3.4.10 Reliability. The design goal of reliability of the M345 shall not be less than 6000 Mean Miles Between Failure (MMBF) of 45% hard-surface roads, 50% secondary roads and 5% cross country operation. For calculation of the MMBF, a failure shall be defined as a condition which prevents operation, or reduces performance below essential levels, or indicates to the crew that further operation would be unsafe, and might result in extensive damage to the equipment. Any of the foregoing conditions which cannot be corrected by the crew in 30 minutes, using the tools and parts normally carried by the M345 or the tow vehicle, constitutes failure. Maintenance and human induced failure are excluded (see 4.5.3.10).

3.4.11 Maintainability. Total maintenance, excluding drivers/crew checks and services, shall not exceed 15 man-hours during 6000 miles(mi) of operation (see 4.5.3.11). This equates to a maintenance ratio of 0.075 at 30 operational mi, equivalent to one operational hour. The schedule maintenance interval shall be 3 months or 3000 mi whichever comes first.

3.5 Safety. All exposed parts which are energized electrical shall be located, insulated, fully enclosed or guarded to prevent hazards to operating personnel and equipment functioning. All moving parts which are of such nature, or so located, as to be a hazard to operating or maintenance personnel shall be enclosed or guarded. Protective devices shall not impair operating functions (see 4.5.4).

3.6 Treatment and painting. Unless otherwise specified (see 6.2), the trailer shall be cleaned, treated, and painted in accordance with normal commercial practice to assure complete coverage and durability of the finish. Unless otherwise specified (see 6.2), the color shall be lusterless forest green (see 4.5.2).

3.7 Identification, marking and data plates. Identification and marking shall be permanent and legible and shall be made of a material suitable to withstand the environmental requirements specified herein (see 4.5.2). The trailer shall be marked with the National Stock Number (NSN) and the manufacturers name and part number.

3.8 Workmanship. The workmanship shall be of a quality to assure that the trailer and its components are free from any defects that compromise, limit or reduce the capability of the trailer in the performance of its intended use (see 4.5.2).

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3.9 Servicing and adjusting. Prior to acceptance by the government, contractor shall adjust and service each trailer for immediate operational use, including at least the following: Adjust brake system, examine electrical system, inflate all tires and completely lubricate the trailer and all running gear with grades of lubricants recommended.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. Unless otherwise specified (see 6.2), first article inspection shall be performed on samples as specified when a first article sample is required (see 3.1). This inspection shall include the examinations of 4.4 (see table I) and the tests of 4.5.3.1 through 4.5.4 (see table II).

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4 (see table I), and the tests of 4.5.3.2, 4.5.3.4, and 4.5.3.5.2 through 4.5.3.9 and 4.5.4 (see table II).

4.4 Examinations.

4.4.1 Sampling. Samples from an inspection lot for conformance inspection shall be selected in accordance with ANSI/ASQC Z1.4. Any redesign or modification of the contractors standard to comply with specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements of requirements as listed in table I. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause of rejection.

4.5 Method of inspection.

4.5.1 Material. Conformance to 3.2, shall be determined by inspection of contractor records providing proof or certification that design, construction, processing and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

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4.5.2 Defects. Conformance to 3.3 through 3.3.4 and 3.6 through 3.9 shall be determined by examination for defects listed in table I. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

Table I. Classification of defects

Category	Defects	Method of inspection
Major:		
101	Dimensions out of tolerance (see 3.3).	SIE 1/
102	Seals not as specified (see 3.3.1).	Visual
103	Electrical system not as specified (see 3.3.2).	Visual
104	Tires and inner tubes not as specified (see 3.3.3).	Visual
105	Adjustable mechanisms not functioning properly (see 3.3.4).	Visual
Minor:		
201	Treatment and Painting not as specified (see 3.6)	Visual
202	Improper identification marking (see 3.7).	Visual
203	Improper workmanship affecting appearance (see 3.8).	Visual
204	Servicing and adjustments not as specified (see 3.9).	Visual

1/ SIE = Standard Inspection Equipment

Table II. Classification of inspections.

Title	Requirements	Inspection	First Article	Quality Conformance
Materials	3.2	4.5.1	X	X
Defects	3.3	4.5.2	X	X
Environmental	3.4.1	4.5.3.1	X	
Road Speeds	3.4.2	4.5.3.2	X	X
Cross-country speeds	3.4.3	4.5.3.3	X	
Grade and slope operation	3.4.4	4.5.3.4	X	X
Service brakes	3.4.5	4.5.3.5	X	
Service brake system	3.4.5.1	4.5.3.5.1	X	
Automatic actuation	3.4.5.2	4.5.3.5.2	X	X
Fording ability	3.4.6	4.5.3.6	X	
Landing legs	3.4.7	4.5.3.7	X	
Turning ability	3.4.8	4.5.3.8	X	
Wheel and tire balancing	3.4.9	4.5.3.9	X	
Reliability	3.4.10	4.5.3.10	X	
Maintainability	3.4.11	4.5.3.11	X	
Safety	3.5	4.5.4	X	

4.5.3 Performance.

4.5.3.1 Environmental. To determine conformance to 3.4.1, the trailer shall be operated and stored at the specified temperature.

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4.5.3.2 Highway. To determine conformance to 3.4.2, the trailer shall be operate on the specified road conditions.

4.5.3.3 Cross country. To determine conformance to 3.4.3, the trailer shall be operate on the specified cross country terrain.

4.5.3.4 Grade and slope operation. To determine conformance to 3.4.4, the trailer shall operate on the specified ascending or descending longitudinal grade and side slopes.

4.5.3.5 Service brake. To determine conformance to 3.4.5, the combination of the service brakes of the trailer and tow vehicle shall stop the combination within the specified distance.

4.5.3.5.1 Service brake system. To determine conformance to 3.4.5.1, the system shall operate to the specified requirements.

4.5.3.5.2 Automatic actuation. To determine conformance to 3.4.5.2, the trailer and tow vehicle shall be held stationary for the specified time.

4.5.3.6 Fording. To determine conformance to 3.4.6, the trailer shall be towed through the water crossing at the depth and for the specified time.

4.5.3.7 Landing leg. To determine conformance to 3.4.7, the landing legs shall operate as specified.

4.5.3.8 Turning ability. To determine conformance to 3.4.8, the trailer and the tow vehicle shall be coupled and shall not interfere with each other while performing the specified turn.

4.5.3.9 Wheel and tire balance. To determine conformance to 3.4.9, each assembly of wheel, tube and tire shall as specified.

4.5.3.10 Reliability. To determine conformance to 3.4.10, reliability shall be verified at a point estimate value equal to or greater than 6000 mi shall be demonstrated during the road test (see table III).

4.5.3.11 Maintainability. To determine conformance to 3.4.11, a maintenance ratio of less than, or equal to, that specified in 3.4.11 must be demonstrated during the road test (see table III).

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Table III. 6000 mile road test.

Course	Mileage and speeds
Hard surfaced roads	2400 mi at varying speeds up to maximum (see 3.4.2).
Gravel and dirt roads	1500 mi at varying speeds up to maximum (see 3.4.2).
Level cross-country	1380 mi at varying speeds up to maximum (see 3.4.3).
Belgian block 1/	120 mi at varying speeds up to maximum
Hilly cross-country	600 mi at speeds applicable to trailer characteristics (see 3.4.3).

1/ = The 6000 mi road test shall include 50 mi of gravel and 30 mi of belgian block without payload at start and completion of highway and cross-country operations, for a total of 160 mi.

4.5.4 Safety. To determine conformance to 3.5, the trailer shall be examined to assure that hazards to operating and maintenance personnel have been eliminated or controlled.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files. CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES.

(This section contains information of a general and explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Trailers covered by this specification are intended for use in hauling items of construction engineering equipment or similar applications during tactical military operations, and for use under extreme or unusual conditions of climate, weather, and terrain.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).
c. When first article samples are required (see 3.1 and 4.2).
d. Treatment and painting, if other than as specified (see 3.6).
e. Packaging requirements (see 5.1).

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6.3 Subject term (key word) listing.

Engineering equipment
Hauling
Recovery
Transporter
Tractors

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army - AT

Preparing Activity:
Army - AT

(Project 2330-0229)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-PRF-45333E(AT)	961129

3. DOCUMENT TITLE
Trailer, Flat Bed: 10 Ton, 4 Wheel M345

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(810) 574-8745	786-8745
c. ADDRESS (Include Zip Code) Commander U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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